

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12396716
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Denham; Gregory J. et al.

Expanding suture anchor having an actuator pin

Abstract

A method and apparatus for reattaching soft tissue to a boney structure using an expanding suture anchor. The method can include providing a suture anchor and forming a cavity in the boney structure for receipt of the suture anchor. The method can also include coupling the soft tissue to a suture and coupling the suture to the suture anchor. The method can further include deploying the suture anchor to fix the suture to the suture anchor and the suture anchor to the cavity.

Inventors: Denham; Gregory J. (Warsaw, IN), Stone; Kevin T. (Winona Lake, IN), Kaiser; Ryan A. (Leesburg, IN), Meridew; Jason D. (Warsaw, IN)

Applicant: Biomet Manufacturing, LLC (Warsaw, IN)

Family ID: 1000008780734

Assignee: Biomet Manufacturing, LLC (Warsaw, IN)

Appl. No.: 17/742927

Filed: May 12, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220265263 A1	Aug. 25, 2022

Related U.S. Application Data

continuation parent-doc US 14665200 20150323 US 11357494 child-doc US 17742927
continuation-in-part parent-doc US 11006398 20041207 US 7976565 child-doc US 11451250
division parent-doc US 11451250 20060612 US 8986345 child-doc US 14665200

Publication Classification

Int. Cl.: A61B17/04 (20060101)

U.S. Cl.:

CPC **A61B17/0401** (20130101); A61B2017/0408 (20130101); A61B2017/0409 (20130101);
A61B2017/0414 (20130101); A61B2017/0422 (20130101); A61B2017/0458 (20130101);
A61B2017/0464 (20130101)

Field of Classification Search

CPC: A61B (17/0401); A61B (17/0487); A61B (2017/0412); A61B (2017/0437); A61B
(17/0469); A61B (17/0446); A61B (17/043); A61B (2017/0414); A61B (2017/0422);
A61B (2017/0409)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3708883	12/1972	Flander	N/A	N/A
4409974	12/1982	Freedland et al.	N/A	N/A
4653489	12/1986	Tronzo	N/A	N/A
4738255	12/1987	Goble et al.	N/A	N/A
5013316	12/1990	Goble et al.	N/A	N/A
5087199	12/1991	Lazarof	N/A	N/A
5167665	12/1991	Mckinney	N/A	N/A
5258016	12/1992	Dipoto et al.	N/A	N/A
5268001	12/1992	Nicholson et al.	N/A	N/A
5326205	12/1993	Anspach, Jr. et al.	N/A	N/A
5464427	12/1994	Curtis et al.	N/A	N/A
5472452	12/1994	Trott	N/A	N/A
5474572	12/1994	Hayhurst	N/A	N/A
5480403	12/1995	Lee et al.	N/A	N/A
5486197	12/1995	Le et al.	N/A	N/A
5501695	12/1995	Anspach, Jr. et al.	N/A	N/A
5531792	12/1995	Huene	N/A	N/A
5584835	12/1995	Greenfield	N/A	N/A
5601558	12/1996	Torrie et al.	N/A	N/A
5643321	12/1996	Mcdevitt	N/A	N/A
5645589	12/1996	Li et al.	N/A	N/A
5649963	12/1996	Mcdevitt	N/A	N/A
5665110	12/1996	Chervitz et al.	N/A	N/A
5713903	12/1997	Sander et al.	N/A	N/A
5720753	12/1997	Sander et al.	N/A	N/A
5725529	12/1997	Nicholson et al.	N/A	N/A
5725541	12/1997	Anspach, III et al.	N/A	N/A
5741282	12/1997	Anspach, III et al.	N/A	N/A
5772672	12/1997	Toy et al.	N/A	N/A
5814071	12/1997	Mcdevitt et al.	N/A	N/A
5911721	12/1998	Nicholson et al.	N/A	N/A

5928244	12/1998	Tovey et al.	N/A	N/A
5931844	12/1998	Thompson et al.	N/A	N/A
5935129	12/1998	Mcdevitt et al.	N/A	N/A
5941901	12/1998	Egan	N/A	N/A
5948000	12/1998	Larsen et al.	N/A	N/A
5948001	12/1998	Larsen	N/A	N/A
5968044	12/1998	Nicholson et al.	N/A	N/A
6022373	12/1999	Li	N/A	N/A
6048343	12/1999	Mathis et al.	N/A	N/A
6056751	12/1999	Fenton, Jr.	N/A	N/A
6129762	12/1999	Li	N/A	N/A
6149669	12/1999	Li	N/A	N/A
6210376	12/2000	Grayson	N/A	N/A
6214012	12/2000	Karpman et al.	N/A	N/A
6227860	12/2000	Hobo	N/A	N/A
6315787	12/2000	Tsugita et al.	N/A	N/A
6319269	12/2000	Li	N/A	N/A
6328758	12/2000	Tornier et al.	N/A	N/A
6447516	12/2001	Bonutti	N/A	N/A
6464706	12/2001	Winters	N/A	N/A
6517564	12/2002	Grafton et al.	N/A	N/A
6524316	12/2002	Nicholson et al.	N/A	N/A
6527794	12/2002	McDevitt	606/1	A61F 2/0811
6554830	12/2002	Chappius	N/A	N/A
6565572	12/2002	Chappius	N/A	N/A
6575976	12/2002	Grafton	N/A	N/A
6582453	12/2002	Tran et al.	N/A	N/A
6599295	12/2002	Tornier et al.	N/A	N/A
6623492	12/2002	Berube et al.	N/A	N/A
6641596	12/2002	Lizardi	N/A	N/A
6645227	12/2002	Fallin et al.	N/A	N/A
6652561	12/2002	Tran	N/A	N/A
6660023	12/2002	Mcdevitt et al.	N/A	N/A
6692516	12/2003	West, Jr. et al.	N/A	N/A
6770073	12/2003	Mcdevitt et al.	N/A	N/A
6840953	12/2004	Martinek	N/A	N/A
6863671	12/2004	Strobel et al.	N/A	N/A
6994725	12/2005	Goble	N/A	N/A
7037324	12/2005	Martinek	N/A	N/A
7250055	12/2006	Vanderwalle	N/A	N/A
7572283	12/2008	Meridew	N/A	N/A
7976565	12/2010	Meridew	N/A	N/A
8057524	12/2010	Meridew	N/A	N/A
8574275	12/2012	Stone et al.	N/A	N/A
8939983	12/2014	Stone et al.	N/A	N/A
8986345	12/2014	Denham et al.	N/A	N/A
2001/0051807	12/2000	Grafton	N/A	N/A
2002/0013608	12/2001	Elattrache et al.	N/A	N/A
2002/0095180	12/2001	Hugh, Jr. et al.	N/A	N/A
2002/0111653	12/2001	Foerster	N/A	N/A

2002/0128684	12/2001	Foerster	N/A	N/A
2002/0147463	12/2001	Martinek	N/A	N/A
2002/0188321	12/2001	Levinson	N/A	N/A
2003/0144667	12/2002	Enayati	N/A	N/A
2003/0181986	12/2002	Buchholz	N/A	N/A
2003/0187444	12/2002	Overaker et al.	N/A	N/A
2004/0133239	12/2003	Singhatat	N/A	N/A
2004/0138706	12/2003	Abrams et al.	N/A	N/A
2004/0138707	12/2003	Greenhalgh	N/A	N/A
2004/0193167	12/2003	Tucciarone et al.	N/A	N/A
2004/0225292	12/2003	Sasso et al.	N/A	N/A
2004/0243179	12/2003	Foerster	N/A	N/A
2004/0267265	12/2003	Kyle	N/A	N/A
2005/0015059	12/2004	Sweeney	N/A	N/A
2005/0075668	12/2004	Lizardi	N/A	N/A
2005/0119698	12/2004	Martinek	N/A	N/A
2005/0137707	12/2004	Malek	N/A	N/A
2006/0004364	12/2005	Green et al.	N/A	N/A
2006/0235413	12/2005	Denham et al.	N/A	N/A
2007/0112352	12/2006	Sorensen et al.	N/A	N/A
2007/0265704	12/2006	Mayer et al.	N/A	N/A
2008/0281325	12/2007	Stone et al.	N/A	N/A
2009/0299386	12/2008	Meridew	N/A	N/A
2011/0004258	12/2010	Stone et al.	N/A	N/A
2014/0066982	12/2013	Stone et al.	N/A	N/A
2015/0134001	12/2014	Stone et al.	N/A	N/A
2015/0257749	12/2014	Denham et al.	N/A	N/A

OTHER PUBLICATIONS

Cannulated ArthroRivet™ Anchor brochure, Arthrotek®, (Aug. 2003). cited by applicant

Cannulated ArthroRivet™, Arthrotek® Inventing the Future of Arthroscopy, [Online]. Retrieved from the Internet: <<http://www.arthrotek.com/prodpage.cfm?c=0A05&p=0102>>, (2005). cited by applicant

“U.S. Appl. No. 11/006,398, Examiner Interview Summary mailed Apr. 5, 2010”, 3 pgs. cited by applicant

“U.S. Appl. No. 11/006,398, Examiner Interview Summary mailed Aug. 24, 2009”, 2 pgs. cited by applicant

“U.S. Appl. No. 11/006,398, Final Office Action mailed Jul. 10, 2009”, 9 pgs. cited by applicant

“U.S. Appl. No. 11/006,398, Final Office Action mailed Aug. 2, 2010”, 9 pgs. cited by applicant

“U.S. Appl. No. 11/006,398, Non Final Office Action mailed Jan. 20, 2010”, 9 pgs. cited by applicant

“U.S. Appl. No. 11/006,398, Non Final Office Action mailed Jan. 29, 2009”, 6 pgs. cited by applicant

“U.S. Appl. No. 11/006,398, Notice of Allowance mailed Mar. 3, 2011”, 11 pgs. cited by applicant

“U.S. Appl. No. 11/006,398, Preliminary Amendment filed Feb. 10, 2006”, 11 pgs. cited by applicant

“U.S. Appl. No. 11/006,398, Response filed Apr. 19, 2010 to Non Final Office Action mailed Jan. 20, 2010”, 21 pgs. cited by applicant

“U.S. Appl. No. 11/006,398, Response filed Apr. 29, 2009 to Non Final Office Action mailed Jan. 29, 2009”, 16 pgs. cited by applicant

“U.S. Appl. No. 11/006,398, Response filed Oct. 12, 2009 to Final Office Action mailed Jul. 10, 2009”, 23 pgs. cited by applicant

“U.S. Appl. No. 11/006,398, Response filed Oct. 20, 2008 to Restriction Requirement mailed Sep. 18, 2008”, 1 pgs. cited by applicant

“U.S. Appl. No. 11/006,398, Response filed Nov. 2, 2010 to Final Office Action mailed Aug. 2, 2010”, 24 pgs. cited by applicant

“U.S. Appl. No. 11/006,398, Restriction Requirement mailed Sep. 18, 2008”, 7 pgs. cited by applicant

“U.S. Appl. No. 11/006,398, Supplemental Amendment filed Jan. 23, 2009”, 11 pgs. cited by applicant

“U.S. Appl. No. 11/006,398, Supplemental Notice of Allowability mailed Apr. 1, 2011”, 2 pgs. cited by applicant

“U.S. Appl. No. 11/451,250, Examiner Interview Summary mailed Jan. 25, 2010”, 3 pgs. cited by applicant

“U.S. Appl. No. 11/451,250, Final Office Action mailed Nov. 20, 2009”, 9 pgs. cited by applicant

“U.S. Appl. No. 11/451,250, Non Final Office Action mailed Apr. 15, 2009”, 6 pgs. cited by applicant

“U.S. Appl. No. 11/451,250, Non Final Office Action mailed Apr. 30, 2014”, 16 pgs. cited by applicant

“U.S. Appl. No. 11/451,250, Notice of Allowance mailed Jan. 12, 2015”, 4 pgs. cited by applicant

“U.S. Appl. No. 11/451,250, Notice of Allowance mailed Nov. 14, 2014”, 9 pgs. cited by applicant

“U.S. Appl. No. 11/451,250, Response filed Feb. 22, 2010 to Final Office Action mailed Nov. 20, 2009”, 16 pgs. cited by applicant

“U.S. Appl. No. 11/451,250, Response filed Mar. 30, 2009 to Restriction Requirement mailed Jan. 30, 2009”, 1 pg. cited by applicant

“U.S. Appl. No. 11/451,250, Response filed Jul. 14, 2009 to Non Final Office Action mailed Apr. 15, 2009”, 16 pgs. cited by applicant

“U.S. Appl. No. 11/451,250, Response filed Jul. 30, 2014 to Non Final Office Action mailed Apr. 30, 2014”, 15 pgs. cited by applicant

“U.S. Appl. No. 11/451,250, Restriction Requirement mailed Jan. 30, 2009”, 7 pgs. cited by applicant

“U.S. Appl. No. 14/665,200, Advisory Action mailed Sep. 8, 2017”, 4 pgs “U.S. Appl. No. 14/665,200, Final Office Action mailed Jan. 7, 2019”, 10 pgs. cited by applicant

“U.S. Appl. No. 14/665,200, Final Office Action mailed Jan. 7, 2019”, 10 pgs. cited by applicant

“U.S. Appl. No. 14/665,200, Final Office Action mailed Jun. 2, 2017”, 12 pgs. cited by applicant

“U.S. Appl. No. 14/665,200, Final Office Action mailed Oct. 5, 2020”, 11 pgs. cited by applicant

“U.S. Appl. No. 14/665,200, Non Final Office Action mailed Mar. 17, 2021”, 10 pgs. cited by applicant

“U.S. Appl. No. 14/665,200, Non Final Office Action mailed Jun. 15, 2018”, 9 pgs. cited by applicant

“U.S. Appl. No. 14/665,200, Non Final Office Action mailed Jul. 8, 2019”, 10 pgs. cited by applicant

“U.S. Appl. No. 14/665,200, Non Final Office Action mailed Aug. 15, 2016”, 13 pgs. cited by applicant

“U.S. Appl. No. 14/665,200, Non Final Office Action mailed Oct. 19, 2017”, 14 pgs. cited by applicant

“U.S. Appl. No. 14/665,200, Non Final Office Action mailed Oct. 29, 2021”, 13 pgs. cited by applicant

“U.S. Appl. No. 14/665,200, Notice of Allowance mailed Feb. 15, 2022”, 7 pgs. cited by applicant

“U.S. Appl. No. 14/665,200, Response filed Feb. 20, 2018 to Non Final Office Action mailed Oct.

2019, 17", 11 pgs. cited by applicant
"U.S. Appl. No. 14/665,200, Response filed Jan. 5, 2021 to Final Office Action mailed Oct. 5, 2020", 10 pgs. cited by applicant
"U.S. Appl. No. 14/665,200, Response filed Jan. 8, 2020 to Non Final Office Action mailed Jul. 8, 2019", 10 pgs. cited by applicant
"U.S. Appl. No. 14/665,200, Response filed Jan. 31, 2022 to Non Final Office Action mailed Oct. 29, 2021", 12 pgs. cited by applicant
"U.S. Appl. No. 14/665,200, Response filed Feb. 15, 2017 to Non Final Office Action mailed Aug. 15, 2016", 10 pgs. cited by applicant
"U.S. Appl. No. 14/665,200, Response filed Apr. 8, 2019 to Final Office Action mailed Jan. 7, 2019", 10 pgs. cited by applicant
"U.S. Appl. No. 14/665,200, Response Filed May 26, 2020 to Notice of Non-Compliant Amendment Mailed Mar. 25, 2020", 10 pgs. cited by applicant
"U.S. Appl. No. 14/665,200, Response filed Jul. 19, 2021 to Non Final Office Action mailed Mar. 17, 2021", 14 pgs. cited by applicant
"U.S. Appl. No. 14/665,200, Response filed Aug. 7, 2017 to Final Office Action mailed Jun. 2, 2017", 11 pgs. cited by applicant
"U.S. Appl. No. 14/665,200, Response filed Aug. 24, 2018 to Non Final Office Actio maled Jun. 15, 2018", 9 pgs. cited by applicant
"U.S. Appl. No. 14/665,200, Response filed Oct. 2, 2017 to Final Office Action mailed Jun. 2, 2017", 11 pgs. cited by applicant
U.S. Appl. No. 11/006,398, filed Dec. 7, 2004, now U.S. Pat. 7,976,565, Expanding Suture Anchor Having an Actuator Pin. cited by applicant
U.S. Appl. No. 11/451,250, filed Jun. 12, 2006, now U.S. Pat. No. 8,986,345, Expanding Suture Anchor Having an Actuator Pin. cited by applicant
U.S. Appl. No. 14/665,200, filed Mar. 23, 2015, Expanding Suture Anchor Having an Actuator Pin. cited by applicant

Primary Examiner: Labranche; Brooke

Assistant Examiner: Lauer; Christina C

Attorney, Agent or Firm: Schwegman Lundberg & Woessner, P.A.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 11/451,250 filed on Jun. 12, 2006, which is a continuation-in-part of U.S. patent application Ser. No. 11/006,398 filed on Dec. 7, 2005, now U.S. Pat. No. 7,976,565 issued on Jul. 12, 2011. The disclosure of the above application is incorporated herein by reference.

FIELD

(1) The present disclosure relates generally to suture anchors, and particularly to a method and apparatus for an expanding suture anchor having an actuator pin.

BACKGROUND

(2) The statements in this section merely provide background information related to the present disclosure and may not constitute prior art.

(3) Various procedures may be performed to repair soft tissue in the body. Generally, it is known to fix the soft tissue to a selected area on the bone by providing a suture through a selected portion of

the soft tissue while securing the other end of the suture to the selected area on the bone using a suture anchor.

(4) Suture anchors may be retained in the selected area of the bone via a feature of the suture anchor. In hard bone, however, the suture anchor may not fully engage the bone because the surgeon is not able to apply sufficient force to the suture. Thus, the suture may become loose in the boney structure, which could lead to increased healing times or improper healing. In addition, some of the techniques used to lock the suture to the suture anchor can be labor intensive. Therefore, it may be desirable to provide a method and apparatus for an expanding suture anchor, and a method and apparatus for locking a suture to such a suture anchor.

SUMMARY

(5) A method for attaching a soft tissue to a boney structure. The method can include providing a suture anchor and forming a cavity in the boney structure for receipt of the suture anchor. The method can also include coupling the soft tissue to a suture and coupling the suture to the suture anchor. The method can further include deploying the suture anchor to fix the suture to the suture anchor and the suture anchor to the cavity.

(6) Also provided is a method for attaching a soft tissue to a boney structure. The method can include providing a suture anchor defining an aperture and forming a cavity in the boney structure for receipt of the suture anchor. The method can further include coupling the soft tissue to a suture and threading at least one end of the suture through the aperture of the suture anchor. The method can include disposing the suture anchor in the cavity and deploying the suture anchor to fix the suture anchor to the cavity. The deployment of the suture anchor can fix the suture to the suture anchor without the use of a knot.

(7) A method for attaching a soft tissue to a boney structure is also provided. The method can include providing a suture anchor and forming a cavity in the boney structure for receipt of the suture anchor. The method can also include coupling the soft tissue to a suture and coupling the suture to the suture anchor. The method can also include disposing the suture anchor in the cavity and applying a retractive force to the suture anchor. The retractive force can fix the suture in the suture anchor.

(8) A suture anchor for attaching soft tissue to a pre-selected area of a boney structure is further provided. The suture anchor can include an expanding anchor for engaging the boney structure and an insert slideably coupled to the expanding anchor. The insert can be moveable to cause the expanding anchor to expand and engage the boney structure. An aperture can be defined in the insert, and a suture can be received in the aperture. The suture can be adapted to be coupled to the soft tissue. The movement of the insert can fix the suture to the expanding anchor.

(9) Further areas of applicability will become apparent from the description provided herein. It should be understood that the description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

DRAWINGS

(1) The drawings described herein are for illustration purposes only and are not intended to limit the scope of the present disclosure in any way.

(2) FIG. 1 is a perspective view of an expanding suture anchor having an actuator pin according to the present disclosure;

(3) FIG. 2 is an exploded view of the suture anchor in FIG. 1;

(4) FIG. 3 is a cross-sectional view taken along line 3-3 of FIG. 1;

(5) FIG. 3A is a detailed view of FIG. 3;

(6) FIG. 4A is an environmental view of a use of the suture anchor shown in FIG. 1;

(7) FIG. 4B is an environmental view of the suture anchor as the retractive force is applied to move the suture anchor into the fully expanded position;

(8) FIG. 4C is an environmental view illustrating the suture anchor in a fully expanded position;

(9) FIG. 4D is an environmental view illustrating an alternate actuator gun for use with the suture anchor shown in FIG. 1;

(10) FIG. 5A is an environmental view illustrating an alternate use of the suture anchor of FIG. 1 after the application of a first retractive force;

(11) FIG. 5B is an environmental view of the suture anchor as a second retractive force is applied;

(12) FIG. 6A is an alternate environmental view of a use of the suture anchor of FIG. 1;

(13) FIG. 6B is an environmental view of the suture anchor of FIG. 6A during the application of a retractive force;

(14) FIG. 7 is a perspective view of a first alternative expanding suture anchor having an actuator pin according to the present disclosure;

(15) FIG. 8 is an environmental view of a first procedure associated with the suture anchor shown in FIG. 7;

(16) FIG. 9 is an environmental view of a second procedure associated with the suture anchor shown in FIG. 7;

(17) FIG. 10 is an environmental view of the suture anchor of FIG. 7 as the retractive force is applied to move the suture anchor into the fully expanded position;

(18) FIG. 11 is an environmental view illustrating the suture anchor of FIG. 7 in a fully expanded position;

(19) FIG. 11A is an alternative environmental partial view illustrating the suture anchor of FIG. 7 in a fully expanded position;

(20) FIG. 12 is detailed front view of the suture anchor of FIG. 7 illustrating a suture fixed to the suture anchor;

(21) FIG. 13 is a perspective view of a second alternative expanding suture anchor having an actuator pin according to the present disclosure;

(22) FIG. 14 is an environmental view of a first procedure associated with the suture anchor shown in FIG. 13;

(23) FIG. 15 is an environmental view of a second procedure associated with the suture anchor shown in FIG. 13;

(24) FIG. 16 is detailed front view of the suture anchor of FIG. 13 illustrating a suture fixed to the suture anchor;

(25) FIG. 17 is a perspective view of a third alternative expanding suture anchor having an actuator pin according to the present disclosure;

(26) FIG. 18 is detailed front view of the suture anchor of FIG. 17 illustrating a suture fixed to the suture anchor;

(27) FIG. 19 is a perspective view of a fourth alternative expanding suture anchor having an actuator pin according to the present disclosure;

(28) FIG. 20 is detailed front view of the suture anchor of FIG. 19 illustrating a suture fixed to the suture anchor;

(29) FIG. 21 is a perspective view of a fifth alternative expanding suture anchor having an actuator pin according to the present disclosure;

(30) FIG. 22 is a cross-sectional view taken along line 22-22 of FIG. 21;

(31) FIG. 23 is an environmental view of a first procedure associated with the suture anchor shown in FIG. 21;

(32) FIG. 24 is an environmental view of a second procedure associated with the suture anchor shown in FIG. 21;

(33) FIG. 25 is an environmental view of the suture anchor of FIG. 21 as the retractive force is applied to move the suture anchor into the fully expanded position;

- (34) FIG. **26** is an environmental view illustrating the suture anchor of FIG. **21** in a fully expanded position;
- (35) FIG. **27** is detailed front view of the suture anchor of FIG. **21** illustrating a suture fixed to the suture anchor;
- (36) FIG. **28** is a perspective view of a sixth alternative expanding suture anchor having an actuator pin according to the present disclosure;
- (37) FIG. **29** is an environmental view of a first procedure associated with the suture anchor shown in FIG. **28**;
- (38) FIG. **30** is an environmental view of a second procedure associated with the suture anchor shown in FIG. **28**;
- (39) FIG. **31** is detailed front view of the suture anchor of FIG. **28** illustrating a suture fixed to the suture anchor;
- (40) FIG. **32** is a perspective view of a seventh alternative expanding suture anchor having an actuator pin according to the present disclosure;
- (41) FIG. **33** is detailed front view of the suture anchor of FIG. **32** illustrating a suture fixed to the suture anchor;
- (42) FIG. **34** is a perspective view of a eighth alternative expanding suture anchor having an actuator pin according to the present disclosure;
- (43) FIG. **35** is detailed front view of the suture anchor of FIG. **34** illustrating a suture fixed to the suture anchor;
- (44) FIG. **36** is a perspective view of a ninth alternative expanding suture anchor having an actuator pin according to the present disclosure;
- (45) FIG. **36A** is another perspective view of a ninth alternative expanding suture anchor having an actuator pin according to the present disclosure;
- (46) FIG. **37** is detailed front view of the suture anchor of FIG. **36** illustrating a suture fixed to the suture anchor;
- (47) FIG. **37B** is another detailed front view of the suture anchor of FIG. **36** illustrating a suture fixed to the suture anchor;
- (48) FIG. **38** is a perspective view of a tenth alternative expanding suture anchor having an actuator pin according to the present disclosure;
- (49) FIG. **39** is detailed front view of the suture anchor of FIG. **38** illustrating a suture fixed to the suture anchor;
- (50) FIG. **40** is a perspective view of a eleventh alternative expanding suture anchor having an actuator pin according to the present disclosure;
- (51) FIG. **41** is an environmental view of a procedure associated with the suture anchor shown in FIG. **40**; and
- (52) FIG. **42** is detailed front view of the suture anchor of FIG. **40** illustrating a suture fixed to the suture anchor.

DETAILED DESCRIPTION

(53) The following description of various embodiments is merely exemplary in nature and is in no way intended to limit the present teachings, their application, or uses. Although the following description is related generally to a suture anchor that can be positioned in a pre-drilled hole, that is a hole provided in a honey structure for acceptance of the suture anchor, it will be understood that a suture anchor including an impacting tip or self-drilling thread may be provided as well. Moreover, it will be understood that the suture anchor, as described and claimed herein, can be used with any appropriate surgical procedure. Therefore, it will be understood that the following discussions is not intended to limit the scope of the appended claims.

(54) With reference to FIG. **1**, an expanding suture anchor **10** having an actuator pin **12** is illustrated. The suture anchor **10** is operable in a first configuration for insertion into a pre-drilled hole **14** in a honey structure **16** (as shown in FIG. **4A**) and operable in a second configuration to

secure the suture anchor **10** in the pre-drilled hole **14** (as shown in FIG. 4C). The suture anchor **10** generally includes an insert **18** molded to the actuator pin **12** and partially disposed in a sleeve **20**. (55) With continuing reference to FIG. 1 and additional reference to FIGS. 2 and 3, the insert **18** includes a suture receiving portion **22** and an end section **24** displaced from the suture receiving portion **22** by a breakaway section **26**. The suture receiving portion **22** and end section **24** have a tensile and a torsion strength that are greater than the tensile and torsion strength of the breakaway section **26** such that the suture receiving portion **22** and end section **24** will be severed by the breakaway section **26** without damage to the structural integrity of either section. The insert **18** is generally insert molded from a resorbable material (such as, for example, Lactosorb®, available from Biomet Inc. of Warsaw, Indiana); however, it will be understood that other types of biocompatible materials and other methods of forming could be used.

(56) At least one groove **28** extends throughout a length L1 of the suture receiving portion **22** and the end section **24** for receipt of strands from a suture **30** therein. The suture receiving portion **22** further has a tip **32** and a cylindrical body **34**. The tip **32** is generally conical in shape; however, any other desired shape may be used. The cylindrical body **34** has a diameter D1, which is sized to ensure a slip fit with the sleeve **20** such that the cylindrical body **34** can slide within the sleeve **20**. The cylindrical body **34** includes a formed eyelet **36** extending through the cylindrical body **34** to provide an attachment point for the suture **30**, as shown in FIG. 3A. The diameter D1 of the cylindrical body can provide an interference fit with regard to the suture **30** fitting between the cylindrical body **34** and the sleeve **20**. Referring back to FIG. 3, the cylindrical body **34** may further include a tapered section **40** leading to the breakaway section **26**.

(57) The breakaway section **26** has a diameter D2, which is generally one-half the size of the diameter D1 of the cylindrical body **34**. The breakaway section **26** may include two necked portions **42**, **44**, which facilitate the fracturing of the breakaway section **26** by reducing the fracture strength of the breakaway section **26** in torsion and tension as compared to the suture receiving portion **22** and the end section **24**. The diameter of the necked portions **42**, **44** may vary to enable different fracture loads for various applications; however, generally the force required to fracture the necked portions **42**, **44** will be between 10-12 foot-pounds. The fracturing of the breakaway section **26** detaches the suture receiving portion **22** from the end section **24** of the insert **18** (as illustrated in FIG. 4C).

(58) With continued reference to FIG. 3 and additional reference to FIG. 2, the end section **24** has a tapered portion **46** coupled to an annular body **48**. The tapered portion **46** includes a first diameter D3 at a first end **50**, which tapers to a second diameter D4 at a second end **52**. In general, the first diameter D3 of the tapered portion **46** is approximately equal to the diameter D1 of the suture receiving portion **22**. The tapered portion **46** operates to place the suture anchor **10** in the second position to engage the boney structure **16** as will be described in greater detail below. The second end **52** of the tapered portion **46** is coupled to the annular body **48**. The annular body **48** is generally cylindrical and has a constant diameter D5. The diameter D5 of the annular body **48** may be larger than or equal to the second diameter D4 of the tapered portion **46**, yet smaller than the first diameter D3 of the tapered portion **46**. The annular body **48** can extend beyond the sleeve **20** when the suture anchor **10** is assembled to provide a locator **54** for an actuator gun **56** (as shown in FIG. 4A) for the application of a retractive force F1; however, this is not necessary, as will be discussed further below. The annular body **48** and tapered portion **46** each further include a formed central bore **58** for receipt of the actuator pin **12** therein.

(59) The actuator pin **12** is fixedly attached to the annular body **48** and tapered portion **46** via insert molding. In particular, the insert **18** is formed around the actuator pin **12**, ensuring secure and precise attachment. The actuator pin **12** is positioned in the formed central bore **58** of the annular body **48** and tapered portion **46** such that the actuator pin **12** is removed from the suture anchor **10** when the breakaway section **26** is fractured. The actuator pin **12** can be made of any suitable biocompatible corrosive resistance material such as, for example, surgical steel. In this regard, the

actuator pin **12** need not be made of the same material as the sleeve **20**.

(60) The actuator pin **12** further includes a formed cavity **60** that retains the actuator pin **12** in the insert **18** through out the application of the retractive force **F1**. More specifically, as best shown in FIG. **3**, as the insert **18** is formed around the actuator pin **12**, material **M** is formed under the cavity **60**. Thus, this material **M** must be displaced in order to remove the actuator pin **12** from the insert **18**. Accordingly, the retractive force **F1** must be less than the fracture strength of the material **M** to ensure the actuator pin **12** is retained in the insert **18**. Similarly, the size of the cavity **60** can be modified to allow the accumulation of varying amounts of material **M** depending on the amount of retractive force **F1** required to secure the suture anchor **10** in a given boney structure.

(61) With continuing reference to FIGS. **2** and **3** and additional reference to FIG. **1**, the sleeve **20** is disposed about a substantial portion of the insert **18**. More specifically, the sleeve **20** includes a throughbore **62** for receipt of the insert **18** therein. The sleeve **20** has a length **L**, which may be configured such that the suture receiving portion **22** and locator **54** are exposed when the suture anchor **10** is in the first position. For example, a first end **64** of sleeve **20** is shown covering the breakaway section **26** and approximately one half of the cylindrical body **34** of the suture receiving portion **22** and a second end **66** of the sleeve **20** covers the tapered portion **46** and approximately one half of the annular body **48** of the end section **24**. The sleeve **20** is typically made from a resorbable material (such as, for example, Lactosorb®, available from Biomet Inc. of Warsaw, Indiana); however, other suitable materials could be employed.

(62) The sleeve **20** is generally cylindrical in nature, and may include at least one ring or securement device **68** on an external surface **70** of the sleeve **20**. Although the sleeve **20** is shown having a ring **68**, it is to be understood various other external features could be disposed on the sleeve **20** (such as, for example, barbs or threads). The at least one ring **68** further aids in the engagement of the suture anchor **10** in the pre-drilled hole **14**. The sleeve **20** also includes at least one expanding member **72** hingably coupled to the sleeve **20** to engage the pre-drilled hole **14**. Specifically, the expanding member **72** is defined by a slot **74** formed in the sleeve **20** and includes a hinge **76**. In one embodiment, the sleeve **20** includes two expanding members **72** located approximately 180 degrees apart. It shall be noted, however, that the expanding members **72** could be located at any position on the sleeve **20** that would engage the boney structure **16**. The expanding members **72** are located at a midsection **78** of the sleeve **20**. The expanding members **72** expand to lock into the boney structure **16** when the retractive force **F1** is applied to the actuator pin **12** and, in turn, to a tapered interior bearing surface **80** on the expanding members **72**.

(63) With reference now to FIG. **4A**, after the pre-drilled hole **14** has been drilled into the boney structure **16** and the eyelet **36** of the insert **18** has been threaded with the suture **30**, the suture anchor **10** can be inserted into the pre-drilled hole **14** as shown. The suture anchor **10** is generally configured such that only the locator **54** of the end section **24** extends beyond a top surface **82** of the boney structure **16**.

(64) Next, the actuator gun **56** is applied to the suture anchor **10** as shown in FIGS. **4A** and **4B**. When a trigger **84** on the actuator gun **56** is pulled, the actuator gun **56** applies the retractive force **F1** to the actuator pin **12**. It should be noted, however, that the actuator gun **56** may not be the only actuating device capable of applying the retractive force **F1** to the suture anchor **10**. In particular, numerous other devices may be employed (such as, for example, a syringe-type actuator **88**, as shown in FIG. **4D**). In FIG. **4D**, the syringe-type actuator **88** includes a cannula **90** that engages the actuator pin **12**. A lever **92** can be used to retract the cannula **90** by the application of a force **F** to the lever **92**. More specifically, the lever **92** operates to retract the cannula **90** once it is engaged with the actuator pin **12** to provide the retractive force **F1** to the suture anchor **10**. A stop **94** can also be provided to prevent the lever **92** from prematurely applying the retractive force **F1**.

(65) The application of the retractive force **F1** causes the insert **18** to displace rearwardly with respect to the sleeve **20**, as illustrated in FIG. **4B**. This rearward displacement causes the tapered portion **46** to apply a force **F3** to the tapered interior bearing surface **80** of the expanding members

72 of the sleeve 20. As the insert 18 continues to move rearward, the tapered portion 46 applies an increasingly greater force F3 to the tapered interior bearing surface 80 of the expanding members 72 until the expanding members 72 are engaged with the boney structure 16.

(66) In a soft boney structure 16, when the expanding members 72 are fully engaged, the tip 32 of the suture receiving portion 22 is pressed against the first end 64 of the sleeve 20, and thus the insert 18 is not able to further retract. Hence, the continued application of the retractive force F1 causes the breakaway section 26 to fracture, as shown in FIG. 40. With reference now to FIG. 5A, in a hard boney structure 16', however, the breakaway section 26 may fracture prior to the tip 32 of the suture receiving portion 22 contacting the first end 64 of the sleeve 20. In this situation, the suture 30 can be pulled by a hand 86 of an operator, as illustrated in FIG. 5B, such that the tip 32 of the suture receiving portion 22 abuts the first end 64 of the sleeve 20. The pulling of the suture 30 by the hand 86 of the operator causes the expanding members 72 to expand further due to the force applied to the tapered interior bearing surface 80 of the expanding members 72 by the cylindrical body 34 of the suture receiving portion 22. In either hard or soft boney structures 16, 16' once the breakaway section 26 fractures, the sleeve 20 and suture receiving portion 22 remain in the pre-drilled hole 14 to couple a selected soft tissue to the boney structures 16, 16' via the suture 30.

(67) In addition, the suture anchor 10 is also adaptable for use in a curved pre-drilled hole 14', as shown in FIG. 6A. In this exemplary embodiment, the suture anchor 10 may be positioned in the curved pre-drilled hole 14' via the actuator pin 12. Next, as shown in FIG. 6B, the retractive force F1 may be applied to the actuator pin 12 by an actuating device, such as the actuator gun 56, to cause the expanding members 72 to engage the boney structure 16'', as discussed previously.

(68) With reference to FIG. 7, a first alternative insert 18a is shown. The first alternative insert 18a can include a suture receiving portion 22a, the breakaway section 26 and the end section 24 for use with a suture anchor 10a substantially similar to that described with regard to FIGS. 1-6B. As will be appreciated, the remainder of the suture anchor 10a can be generally similar to the suture anchor 10 that is illustrated in and described in conjunction with FIGS. 1-6B. In addition, as the breakaway section 26 and end section 24 of the first alternative insert 18a are substantially similar to the breakaway section 26 and end section 24 of the insert 18 of the suture anchor 10, the breakaway section 26 and end section 24 will not be discussed in detail with regard to the first alternative insert 18a. Further, the same reference numerals will be used to denote the same or similar components.

(69) The suture receiving portion 22a and end section 24 can have a tensile and a torsion strength that are greater than the tensile and torsion strength of the breakaway section 26 such that the suture receiving portion 22a and end section 24 will be severed by the breakaway section 26 without damage to the structural integrity of either section. The first alternative insert 18a is generally insert molded from a resorbable material (such as, for example, Lactosorb® available from Biomet Inc. of Warsaw, Indiana); however, it will be understood that other types of biocompatible materials and other methods of forming could be used.

(70) As shown in FIG. 7, the suture receiving portion 22a can include the tip 32, a cylindrical body 34a, and a suture 30a. The tip 32 can be generally conical in shape; however, any other desired shape may be used, such as rectangular. The cylindrical body 34a can have a diameter D1a, which is sized to ensure a slip fit with the sleeve 20 such that the cylindrical body 34a can slide within the sleeve 20. The cylindrical body 34a can define an aperture, which can be an eyelet 36a, extending through the cylindrical body 34a to provide an attachment point for the suture 30a. The diameter D1a of the cylindrical body 34a can also provide an interference fit between the cylindrical body 34a, the suture 30a and the sleeve 20. The cylindrical body 34a can further include the tapered section 40 leading to the breakaway section 26. The suture 30a can have a first end 200, a midsection 201 and a second end 202. The eyelet 36a can be sized to receive both the first end 200 and the second end 202 of the suture 30a. The suture 30a can be fixed to the cylindrical body 34a without the use of a knot by the sleeve 20, as will be discussed herein. The midsection 201 of the

suture **30a** can be coupled to a section of soft tissue **204**, as shown in FIG. 8. The midsection **201** of the suture **30a** can be coupled to the soft tissue **204** through any appropriate technique, such as a mattress stitch **206** or by using a suture punch (not shown).

(71) In order to employ the suture anchor **10a**, with the pre-drilled hole **14** drilled into the boney structure **16**, the suture **30a** can be coupled to the soft tissue **204**. Then, the hand **86** of the operator can thread or insert the first end **200** and the second end **202** of the suture **30a** through the eyelet **36a**. With additional reference to FIG. 9, once the first end **200** and the second end **202** of the suture **30a** are threaded through the eyelet **36a**, the suture anchor **10a** can be inserted into the pre-drilled hole **14**. Generally, the first end **200** and the second end **202** of the suture **30a** can extend beyond the pre-drilled hole **14** to enable the hand **86** of the operator to grasp either or both the first end **200** and the second end **202**. The hand **86** of the operator can be used to tighten the suture **30a**, and thus the soft tissue **204**, to the pre-drilled hole **14** by pulling the first end **200** and second end **202** of the suture **30a** that extends from the pre-drilled hole **14**. This further removes any slack from the suture **30a**. After the suture **30a** is tightened, the expanding members **72** of the suture anchor **10a** can be expanded to engage the pre-drilled hole **14**.

(72) For example, in order to expand the expanding members **72**, an alternative syringe-type actuator **88'** can be coupled to the actuator pin **12** of the suture anchor **10a** as shown in FIGS. 9 and 10. The alternative syringe-type actuator **88'** can include a cannula **90'** that engages the actuator pin **12**. A lever **92'** can be used to retract the cannula **90'** by the application of a force **F** to the lever **92'**. Generally, the lever **92'** can operate to retract the cannula **90'** once it is engaged with the actuator pin **12** to provide the retractive force **F1** to the suture anchor **10a**. A safety or stop **94'** can also be provided to prevent the operator from prematurely applying the retractive force **F1**.

(73) In order to deploy the suture anchor **10a** in the pre-drilled hole **14**, the stop **94'** can be pushed in and then the force **F** can be applied to the lever **92'** on the syringe-type actuator **88'**, as shown in FIG. 10. When the lever **92'** on the alternative syringe-type actuator **88'** is pulled, the alternative syringe-type actuator **88'** can apply the retractive force **F1** to the actuator pin **12** via the cannula **90'**. It should be noted, however, that the alternative syringe-type actuator **88'** is not the only actuating device capable of applying the retractive force **F1** to the suture anchor **10a**. Numerous other devices can be employed, such as, a syringe-type actuator **88**, as discussed with regard to FIG. 4D, or the actuator gun **56** discussed with regard to FIGS. 4A and 4B.

(74) The application of the retractive force **F1** can cause the first alternative insert **18a** to be displaced or moved rearwardly with respect to the sleeve **20**, as illustrated in FIG. 10. This rearward movement can cause the tapered portion **46** of the first alternative insert **18a** to apply a force **F3** to the tapered interior bearing surface **80** of the expanding members **72** of the sleeve **20**. The first alternative insert **18a** can continue to move rearward, which can cause the tapered portion **46** to apply an increasingly greater force **F3** to the tapered interior bearing surface **80** of the expanding members **72** until the expanding members **72** are engaged with the boney structure **16**.

(75) When the expanding members **72** are fully engaged, the tip **32** of the suture receiving portion **22a** can be adjacent to the suture **30a**, while the suture **30a** is adjacent to the first end **64** of the sleeve **20** (as best shown in FIG. 12). The compression of the suture **30a** between the suture receiving portion **22a** and the tip **32** can fix or lock the suture **30a** to the suture anchor **10a**, while preventing the first alternative insert **18a** from retracting further. The continued application of the retractive force **F1** to the first alternative insert **18a** can cause the breakaway section **26** to fracture, as shown in FIG. 11.

(76) It should be noted, as shown in FIG. 11A, the pre-drilled hole **14** can be sized with a depth **D** substantially equivalent to a length **L** of the suture anchor **10a**, if desired. If the pre-drilled hole **14** has the depth **D** substantially equivalent to the length **L** of the suture anchor **10a**, then the expanding members **72** can be deployed or expanded by the contact of the tip **32** of the suture receiving portion **22a** against a bottom **213** of the pre-drilled hole **14**. Generally, as the suture anchor **10a** is inserted, the tip **32** can be pushed against the bottom **213** of the pre-drilled hole **14**,

forcing the first alternative insert **18a** to be displaced or moved rearwardly with respect to the sleeve **20**. This rearward movement can cause the tapered portion **46** of the first alternative insert **18a** to apply a force **F3** to the tapered interior bearing surface **80** of the expanding members **72** of the sleeve **20**. Then, the lever **92'** on the syringe-type actuator **88'** can be pulled (with the stop **94'** pushed in) to apply the retractive force **F1** to the actuator pin **12** to fully engage the expanding members **72** in the pre-drilled hole **14**. The continued application of the retractive force **F1** to the first alternative insert **18a** can cause the breakaway section **26** to fracture, as shown in FIG. **11**.

(77) Once the breakaway section **26** fractures, as shown in either FIG. **11** or **11A**, the sleeve **20** and suture receiving portion **22a** remain in the pre-drilled hole **14** to couple the soft tissue **204** to the boney structure **16** via the suture **30a**. The first end **200** and second end **202** of the suture **30a** that extends from the pre-drilled hole **14** can then be trimmed if desired (not shown).

(78) As shown in FIG. **12**, the suture anchor **10a** can fix the suture **30a** to the suture anchor **10a** without the use of a knot. Generally, as the first end **200** and second end **202** of the suture **30a** extend from the eyelet **36a** of the first alternative insert **18a** prior to the deployment of the expanding members **72** of the sleeve **20**, the application of the retractive force **F1** causes the first alternative insert **18a**, and thus suture **30a**, to be withdrawn into the sleeve **20**. Thus, the movement of the first alternative insert **18a** within the sleeve **20** can fix or lock the suture **30a** to the suture anchor **10a** due to the interference fit between the first alternative insert **18a**, the suture **30a** and the sleeve **20**. Further, the interference fit between the first alternative insert **18a**, the suture **30a** and the sleeve **20** can prevent the first alternative insert **18a** from moving after the deployment of the expanding members **72** due to the frictional lock created between the suture **30a**, the first alternative insert **18a** and the sleeve **20**.

(79) With reference to FIG. **13**, a second alternative insert **18b** is shown. The second alternative insert **18b** can include a suture receiving portion **22b**, the breakaway section **26**, and the end section **24** for use with a suture anchor **10b** substantially similar to that described with regard to FIGS. **7-12**. As the breakaway section **26** and end section **24** of the second alternative insert **18b** are substantially similar to the breakaway section **26** and end section **24** of the insert **18**, the breakaway section **26** and end section **24** will not be discussed in detail with regard to the second alternative insert **18b**. In addition, as the suture receiving portion **22b** is substantially similar to the suture receiving portion **22a** of the first alternative insert **18a**, as discussed with FIGS. **7-12**, only the modifications to the suture receiving portion **22b** will be discussed herein.

(80) As shown in FIG. **13**, the suture receiving portion **22b** can include an aperture, for example, a groove or slot **250**, which can be for receipt of the suture **30a**. Generally, the slot **250** can be formed longitudinally in a cylindrical body **34b** of the second alternative insert **18b**; however, the slot **250** can be formed in any desired position, such as vertically or diagonally. As will be appreciated, the remainder of the cylindrical body **34b** can be generally similar to the cylindrical body **34a** that is illustrated in and described in conjunction with FIG. **7**.

(81) The slot **250** can extend for a length **L** and can have a height **H** configured to enable an operator to insert the first end **200** and second end **202** of the suture **30a** through the slot **250**, as shown in FIG. **14**. Generally, once the midsection **201** of the suture **30a** is coupled to the soft tissue **204** via the mattress stitch **206** or any other technique, such as a suture punch (not shown), the hand **86** of the operator can thread or insert the first end **200** and second end **202** of the suture **30a** into the slot **250**. Then, with reference to FIG. **15**, the suture anchor **10b** can be inserted into the pre-drilled hole **14**. As discussed previously, the hand **86** of the operator can be used to tighten the suture **30a**, and thus the soft tissue **204**, to the pre-drilled hole **14** by pulling the first end **200** and second end **202** of the suture **30a** that extends from the pre-drilled hole **14**. After the suture **30a** is tightened, the suture anchor **10b** can be deployed to engage the pre-drilled hole **14** through the use of the actuator gun **56**, syringe-type actuator **88** or syringe-type actuator **88'**, for example, as discussed previously with reference to FIGS. **4A** and **4B**, **4D** and **9-12**, respectively.

(82) When the suture anchor **10b** is fully engaged to the pre-drilled hole **14**, the tip **32** of the suture

receiving portion **22b** can be adjacent to the suture **30a**, while the suture **30a** is adjacent to the first end **64** of the sleeve **20** (as best shown in FIG. **16**). The compression of the suture **30a** between the suture receiving portion **22b** and the tip **32** can fix or lock the suture **30a** to the suture anchor **10b**. Thus, the suture anchor **10b** can also fix the suture **30a** to the suture anchor **10b** without the use of a knot, while providing greater access for the hand **86** of the operator to couple the first end **200** and the second end **202** to the second alternative insert **18b** prior to the placement of the suture anchor **10b** into the pre-drilled hole **14**. The first end **200** and second end **202** of the suture **30a** that extends from the pre-drilled hole **14** can then be trimmed if desired (not shown).

(83) With reference to FIG. **17**, a third alternative insert **18c** is shown. The third alternative insert **18c** can include a suture receiving portion **22c**, the breakaway section **26**, and the end section **24** for use with a suture anchor **10c** substantially similar to that described with regard to FIGS. **7-12**. As the breakaway section **26** and end section **24** of the third alternative insert **18c** are substantially similar to the breakaway section **26** and end section **24** of the insert **18**, the breakaway section **26** and end section **24** will not be discussed in detail with regard to the third alternative insert **18c**. In addition, as the suture receiving portion **22c** is substantially similar to the suture receiving portion **22a** of the first alternative insert **18a**, as discussed with FIGS. **7-12**, only the modifications to the suture receiving portion **22c** will be discussed herein.

(84) As shown in FIG. **17**, the suture receiving portion **22c** can include a slot, for example, a C-shaped slot **260**, which can be for receipt of the suture **30a**. Generally, the slot **260** can be defined in a cylindrical body **34c** of the third alternative insert **18c**. As will be appreciated, the remainder of the cylindrical body **34c** can be generally similar to the cylindrical body **34a** that is illustrated in and described in conjunction with FIG. **7**. The slot **260** can define a throughbore **262** and a necked portion **264** as shown in FIGS. **17** and **18**. The necked portion **264** can define an aperture **266** for receipt of the first end **200** and second end **202** of the suture **30a**. Generally, the necked portion **264** can be configured such that the first end **200** and second end **202** of the suture **30a** can be hooked behind and retained by the necked portion **264**.

(85) As the assembly and deployment of the suture anchor **10c** can be substantially similar to the assembly and deployment of the suture anchor **10a** discussed with regard to FIGS. **8-12** and the suture anchor **10b** discussed with regard to FIGS. **15** and **16**, the assembly and deployment of the suture anchor **10c** will not be discussed in detail herein. Briefly, however, once the midsection **201** of the suture **30a** is coupled to the soft tissue **204** via the mattress stitch **206** or any other technique, such as a suture punch (not shown), the hand **86** of the operator can thread or insert the first end **200** and second end **202** of the suture **30a** through the aperture **266** of the necked portion **264** and into the throughbore **262** (not shown). Then, the suture anchor **10c** can be inserted into the pre-drilled hole **14**. The hand **86** of the operator can be used to tighten the suture **30a**, and thus the soft tissue **204**, to the pre-drilled hole **14** by pulling the first end **200** and second end **202** of the suture **30a** that extends from the pre-drilled hole **14** (not shown). This further removes any slack from the suture **30a**. After the suture **30a** is tightened, the expanding members **72** of the suture anchor **10c** can be deployed or expanded to engage the pre-drilled hole **14**.

(86) When the suture anchor **10c** is fully engaged to the pre-drilled hole **14**, the tip **32** of the suture receiving portion **22c** can be adjacent to the suture **30a**, while the suture **30a** is adjacent to the first end **64** of the sleeve **20** (as best shown in FIG. **18**). The compression of the suture **30a** between the suture receiving portion **22c** and the tip **32** can fix or lock the suture **30a** to the suture anchor **10c**. Thus, the suture anchor **10c** can also fix the suture **30a** to the suture anchor **10c** without the use of a knot. The first end **200** and second end **202** of the suture **30a** that extends from the pre-drilled hole **14** can then be trimmed if desired (not shown).

(87) With reference to FIG. **19**, a fourth alternative insert **18d** is shown. The fourth alternative insert **18d** can include a suture receiving portion **22d**, the breakaway section **26**, and the end section **24** for use with a suture anchor **10d** substantially similar to that described with regard to FIGS. **7-12**. As the breakaway section **26** and end section **24** of the fourth alternative insert **18d** are

substantially similar to the breakaway section **26** and end section **24** of the insert **18**, the breakaway section **26** and end section **24** will not be discussed in detail with regard to the fourth alternative insert **18d**. In addition, as the suture receiving portion **22d** is substantially similar to the suture receiving portion **22a** of the first alternative insert **18a**, as discussed with FIGS. 7-12, only the modifications to the suture receiving portion **22d** will be discussed herein.

(88) As shown in FIG. 19, the suture receiving portion **22d** can include a hinged slot **270** which can be for receipt of the suture **30a**. Generally, the hinged slot **270** can be defined in a cylindrical body **34d** of the fourth alternative insert **18d**. As will be appreciated, the remainder of the cylindrical body **34d** can be generally similar to the cylindrical body **34a** that is illustrated in and described in conjunction with FIG. 7.

(89) The hinged slot **270** can define a throughbore **272** and can include a hinged portion **274**. The hinged portion **274** can optionally include at least one or a plurality of barbs or projections **276**. The throughbore **272** can be configured to retain the first end **200** and second end **202** of the suture **30a**. The hinged portion **274** can be pivotably coupled to the cylindrical body **34d** and can rotate from an opened position to a closed position. In the opened position, as shown in FIG. 19, the hinged portion **274** can define an aperture **278** for receipt of the first end **200** and second end **202** of the suture **30a** to enable the suture **30a** to enter the throughbore **272** of the hinged slot **270**. In the closed position, the hinged portion **274** can seal or enclose the aperture **278** to retain the suture **30a** within the throughbore **272**, as shown in FIG. 20. Thus, generally, the hinged portion **274** can be sized such that in the closed position the hinged portion **274** can substantially enclose the aperture **278**. The projections **276** of the hinged portion **274** are optional and can be coupled to or formed with the hinged portion **274** if desired. The projections **276** can extend into the throughbore **272** when the hinged portion **274** is in the closed position to further retain the suture **30a** within the throughbore **272**. The projections **276** are illustrated as triangular, however, any appropriate shape, such as annular or hooked, could be employed.

(90) As the assembly and deployment of the suture anchor **10d** in the pre-drilled hole **14** is substantially similar to the assembly and deployment of the suture anchor **10a** discussed with regard to FIGS. 8-12 and the suture anchor **10b** discussed with regard to FIGS. 15 and 16, the assembly and deployment of the suture anchor **10d** will not be discussed in detail herein. Briefly, however, once the midsection **201** of the suture **30a** is coupled to the soft tissue **204** via the mattress stitch **206** or any other technique, such as a suture punch, the hand **86** of the operator can thread or insert the first end **200** and second end **202** of the suture **30a** through the aperture **278** of the hinged portion **274** and into the throughbore **272** (not shown). Then, the suture anchor **10d** can be inserted into the pre-drilled hole **14** (not shown). The hand **86** of the operator can be used to tighten the suture **30a**, and thus the soft tissue **204**, to the pre-drilled hole **14** by pulling the first end **200** and second end **202** of the suture **30a** that extends from the pre-drilled hole **14** (not shown). This further removes any slack from the suture **30a**. After the suture **30a** is tightened, the expanding members **72** of the suture anchor **10d** can be deployed to engage the pre-drilled hole **14**. During the deployment of the suture anchor **10d**, as the fourth alternative insert **18d** moves rearwardly in the sleeve **20**, the hinged portion **274** can be pushed closed by the sleeve **20** to lock the suture **30a** within the throughbore **272** of the suture receiving portion **22d**.

(91) When the suture anchor **10d** is fully engaged to the pre-drilled hole **14**, the tip **32** of the suture receiving portion **22d** can be adjacent to the suture **30a**, while the suture **30a** is adjacent to the first end **64** of the sleeve **20** (as best shown in FIG. 20). The compression of the suture **30a** between the suture receiving portion **22d** and the tip **32** can fix or lock the suture **30a** to the suture anchor **10d**. Thus, the suture anchor **10d** can also fix the suture **30a** to the suture anchor **10d** without the use of a knot. The first end **200** and second end **202** of the suture **30a** that extends from the pre-drilled hole **14** can then be trimmed if desired (not shown).

(92) With reference to FIG. 21, a fifth alternative insert **18e** is shown. The fifth alternative insert **18e** can include a suture receiving portion **22e**, the breakaway section **26**, and the end section **24**

for use with a suture anchor **10e** substantially similar to that described with regard to FIGS. **1-6B**. As will be appreciated, the remainder of the suture anchor **10e** can be generally similar to the suture anchor **10** that is illustrated in and described in conjunction with FIGS. **1-6B**. As the breakaway section **26** and end section **24** of the fifth alternative insert **18e** are substantially similar to the breakaway section **26** and end section **24** of the insert **18**, the breakaway section **26** and end section **24** will not be discussed in detail with regard to the fifth alternative insert **18e**.

(93) The suture receiving portion **22e** and end section **24** can have a tensile and a torsion strength that are greater than the tensile and torsion strength of the breakaway section **26** such that the suture receiving portion **22e** and end section **24** will be severed by the breakaway section **26** without damage to the structural integrity of either section. The fifth alternative insert **18e** is generally insert molded from a resorbable material (such as, for example, Lactosorb® available from Biomet Inc. of Warsaw, Indiana); however, it will be understood that other types of biocompatible materials and other methods of forming could be used.

(94) As shown in FIG. **21**, the suture receiving portion **22e** can include the tip **32**, a cylindrical body **34e** and a suture **30e**. The tip **32** can be generally conical in shape; however, any other desired shape may be used, such as rectangular. The cylindrical body **34e** can have a diameter **D1e**, which is sized to ensure a slip fit with the sleeve **20** such that the cylindrical body **34e** can slide within the sleeve **20**. The cylindrical body **34e** can define an aperture or eyelet **36e** extending through the cylindrical body **34e** to provide an attachment point for the suture **30e**. The cylindrical body **34e** can further include the tapered section **40** leading to the breakaway section **26**.

(95) With additional reference to FIGS. **22** and **23**, the suture **30e** can have a first end **300**, a midsection **302** and a second end **304**. The first end **300** of the suture **30e** can be coupled to the breakaway section **26** of the fifth alternative insert **18e**. Generally, the first end **300** can be knotted about the breakaway section **26** at a midpoint **307** of the two necked portions **42**, **44** (FIG. **22**). The midsection **302** of the suture **30e** can be coupled to a section of soft tissue **306** (FIG. **23**). The second end **304** of the suture **30e** can be received into the eyelet **36e** of the cylindrical body portion **34e** of the fifth alternative insert **18e**. The second end **304** can be coupled to a needle **309** (shown in phantom) to facilitate the threading of the second end **304** into the eyelet **36e** (FIG. **22**). The second end **304**, once threaded, can pass through the eyelet **36e** to secure the soft tissue **306** to the pre-drilled hole **14**, as will be discussed in greater detail herein.

(96) With reference to FIG. **23**, prior to the assembly of the sleeve **20** with the fifth alternative insert **18e**, the first end **300** of the suture **30e** can be knotted to the midpoint **307** of the breakaway section **26** of the fifth alternative insert **18e**. Then, the sleeve **20** can be disposed over the breakaway section **26** of the fifth alternative insert **18e**, such that the second end **304** of the suture **30e** can extend out from the sleeve **20**. Then, with the pre-drilled hole **14** drilled into the honey structure **16**, and the first end the suture **30e** coupled to the breakaway section **26**, the midsection **302** of the suture **30e** can be coupled to the soft tissue **306**. The suture **30e** can be coupled to the soft tissue **306** through any appropriate technique, such as a mattress stitch **308** or a suture punch (not shown), for example. Then, the hand **86** of an operator can thread or insert the second end **304** of the suture **30e** through the eyelet **36e** with or without the use of the needle **309**.

(97) With additional reference to FIG. **24**, once the second end **304** of the suture **30e** is threaded through the eyelet **36e**, the suture anchor **10e** can be inserted into the pre-drilled hole **14**. Generally, the second end **304** of the suture **30e** can extend beyond the pre-drilled hole **14** to enable the hand **86** of the operator to grasp the second end **304** to tighten the suture **30e**, and thus the soft tissue **306**, to the pre-drilled hole **14**. Typically, the suture **30e** can be tightened by pulling the second end **304** of the suture **30e** that extends beyond the pre-drilled hole **14**. This further removes any slack from the suture **30e**. After the suture **30e** is tightened, the expanding members **72** of the suture anchor **10e** can be deployed to engage the pre-drilled hole **14**.

(98) As discussed herein, the alternative syringe-type actuator **88'** can be used to expand the expanding members **72** of the suture anchor **10e**; however, any other activating device, such as the

actuator gun **56** discussed with regard to FIGS. **4A** and **4B**, or the syringe-type actuator **88** discussed with regard to FIG. **4D**, could be used. If the alternative syringe-type actuator **88'** is used, then the lever **92'** on the alternative syringe-type actuator **88'** can be pulled, as shown in FIG. **25**, to apply the retractive force **F1** to the actuator pin **12**. The application of the retractive force **F1** can cause the fifth alternative insert **18e** to be displaced or moved rearwardly with respect to the sleeve **20** and can cause the tapered portion **46** of the fifth alternative insert **18e** to apply a force **F3** to the tapered interior bearing surface **80** of the expanding members **72** of the sleeve **20** until the expanding members **72** are engaged with the boney structure **16**.

(99) When the expanding members **72** are fully engaged, the tip **32** of the suture receiving portion **22e** can be adjacent to the suture **30e**, while the suture **30e** is adjacent to the first end **64** of the sleeve **20** (as best shown in FIG. **19**). The compression of the suture **30e** between the suture receiving portion **22e** and the tip **32** can fix or lock the suture **30e** to the suture anchor **10e**, while preventing the fifth alternative insert **18e** from retracting further. The continued application of the retractive force **F1** to the fifth alternative insert **18e** can cause the breakaway section **26** to fracture, as shown in FIG. **26**. Once the breakaway section **26** fractures, the sleeve **20** and suture receiving portion **22e** remain in the pre-drilled hole **14** to couple the soft tissue **306** to the honey structure **16** via the suture **30e**. The portion of the second end **304** that extends from the pre-drilled hole **14** can then be trimmed if desired (not shown).

(100) As shown in FIG. **27**, the suture anchor **10e** can fix or lock the second end **304** of the suture **30e** to the suture anchor **10e** without the use of a knot. In addition, the suture anchor **10e** can ensure the first end **300** of the suture **30e** remains fixed to the fifth alternative insert **18e**. Generally, as a portion of the first end **300** extends from the sleeve **20** and the second end **304** extends from the eyelet **36e** of the fifth alternative insert **18e** prior to the deployment of the expanding members **72** of the sleeve **20**, the application of the retractive force **F1** causes the fifth alternative insert **18e**, and thus the portion of the first end **300** and the second end **304** extending from the eyelet **36e**, to be withdrawn into the sleeve **20**. Thus, the movement of the fifth alternative insert **18e** within the sleeve **20** can fix or lock the portion of the first end **300** and the second end **304** to the suture anchor **10e** due to the interference fit between the fifth alternative insert **18e**, the suture **30e**, and the sleeve **20**, as discussed previously. Additionally, the frictional lock created between the suture **30e**, the fifth alternative insert **18e** and the sleeve **20** can prevent the fifth alternative insert **18e** from moving after the deployment of the expanding members.

(101) With reference to FIG. **28**, a sixth alternative insert **18f** is shown. The sixth alternative insert **18f** can include a suture receiving portion **22f**, the breakaway section **26**, and the end section **24** for use with a suture anchor **10f** substantially similar to the suture anchor **10e** described with regard to FIGS. **21-27**. As the breakaway section **26** and end section **24** of the sixth alternative insert **18f** are substantially similar to the breakaway section **26** and end section **24** of the insert **18**, the breakaway section **26** and end section **24** will not be discussed in detail with regard to the sixth alternative insert **18f**. In addition, as the suture receiving portion **22f** is substantially similar to the suture receiving portion **22e** of the fifth alternative insert **18e**, as discussed with reference to FIGS. **21-27**, only the modifications to the suture receiving portion **22f** will be discussed herein.

(102) As shown in FIG. **28**, the suture receiving portion **22f** can include an aperture, for example, a groove or slot **350**, which can be for receipt of the suture **30e**. Generally, the slot **350** can be formed longitudinally in a cylindrical body **34f** of the sixth alternative insert **18f**; however, the slot **350** can be formed in any desired position, such as vertically or diagonally. As will be appreciated, the remainder of the cylindrical body **34f** can be generally similar to the cylindrical body **34e** that is illustrated in and described in conjunction with FIG. **21**.

(103) The slot **350** can extend for a length **L** and can have a height **H** configured to enable an operator to insert the second end **304** of the suture **30e** through the slot **350**, as shown in FIG. **29**. Generally, once the first end **300** of the suture **30e** is tied to the breakaway section **26** and the midsection **302** of the suture **30e** is coupled to the soft tissue **306** via the mattress stitch **308** or any

other technique, such as a suture punch (not shown), the hand **86** of the operator can thread or insert the second end **304** of the suture **30e** into the slot **350**.

(104) With reference to FIG. **30**, the suture anchor **10f** can be inserted into the pre-drilled hole **14** formed in the boney structure **16**. Next, as discussed herein, the hand **86** of the operator can be used to tighten the suture **30e**, and thus the soft tissue **306**, to the pre-drilled hole **14** by pulling the second end **304** of the suture **30e** that extends from the pre-drilled hole **14**. After the suture **30e** is tightened, the expanding members **72** of the suture anchor **10f** can be expanded to engage the pre-drilled hole **14** through the use of the actuator gun **56**, syringe-type actuator **88** or syringe-type actuator **88'**, for example, as discussed previously with reference to FIGS. **4A** and **4B**, **4D** and **9-12**, respectively.

(105) When the suture anchor **10f** is fully engaged to the pre-drilled hole **14**, the tip **32** of the suture receiving portion **22f** can be adjacent to the suture **30e**, while the suture **30e** is adjacent to the first end **64** of the sleeve **20**, as best shown in FIG. **31**. The compression of the suture **30e** between the suture receiving portion **22f** and the tip **32** can fix or lock the suture **30e** to the suture anchor **10f**. Thus, the suture anchor **10f** can fix the second end **304** of the suture **30e** to the suture anchor **10f** without the use of a knot, while providing greater access for the operator to couple the second end **304** of the suture **30e** to the sixth alternative insert **18f** prior to the insertion of the suture anchor **10f** into the pre-drilled hole **14**. The portion of the second end **304** that extends from the pre-drilled hole **14** can then be trimmed if desired (not shown).

(106) With reference to FIG. **32**, a seventh alternative insert **18g** is shown. The seventh alternative insert **18g** can include a suture receiving portion **22g**, the breakaway section **26**, and the end section **24** for use with a suture anchor **10g** substantially similar to the suture anchor **10e** described with regard to FIGS. **21-27**. As the breakaway section **26** and end section **24** of the seventh alternative insert **18g** are substantially similar to the breakaway section **26** and end section **24** of the insert **18**, the breakaway section **26** and end section **24** will not be discussed in detail with regard to the seventh alternative insert **18g**. In addition, as the suture receiving portion **22g** is substantially similar to the suture receiving portion **22e** of the fifth alternative insert **18e**, as discussed with FIGS. **21-27**, only the modifications to the suture receiving portion **22g** will be discussed herein.

(107) As shown in FIG. **32**, the suture receiving portion **22g** can include a slot, for example, a C-shaped slot **360**, which can be for receipt of the suture **30e**. Generally, the slot **360** can be defined in a cylindrical body **34g** of the seventh alternative insert **18g**. As will be appreciated, the remainder of the cylindrical body **34g** can be generally similar to the cylindrical body **34e** that is illustrated in and described in conjunction with FIG. **21**. The slot **360** can define a throughbore **362** and a necked portion **364**. The necked portion **364** can define an aperture **366** for receipt of the first end **300** and second end **304** of the suture **30e**. Generally, the necked portion **364** can be configured such that the first end **300** and second end **304** of the suture **30e** can be hooked behind and retained by the necked portion **364**.

(108) As the assembly and deployment of the suture anchor **10g** in the pre-drilled hole **14** is substantially similar to the deployment of the suture anchor **10e** discussed with regard to FIGS. **21-27** and the suture anchor **10f** discussed with regard to FIGS. **30** and **31**, the assembly and deployment of the suture anchor **10g** will not be discussed in detail herein. Briefly, however, the first end **300** of the suture **30e** can be tied to the breakaway section **26** and the midsection **302** of the suture **30e** can be coupled to the soft tissue **306** via the mattress stitch **308** or any other technique, such as a suture punch (not shown). Then, the hand **86** of the operator can thread or insert the second end **304** of the suture **30e** into the slot **360** (not shown). Then, the suture anchor **10g** can be inserted into the pre-drilled hole **14** and the hand **86** of the operator can be used to tighten the suture **30e**, and thus the soft tissue **306**, to the pre-drilled hole **14** by pulling the second end **304** of the suture **30e** that extends from the pre-drilled hole **14** (not shown). This further removes any slack from the suture **30e**. After the suture **30e** is tightened, the expanding members **72** of the suture anchor **10g** can be deployed to engage the pre-drilled hole **14**.

(109) When the suture anchor **10g** is fully engaged to the pre-drilled hole **14**, the tip **32** of the suture receiving portion **22g** can be adjacent to the suture **30e**, while the suture **30e** is adjacent to the first end **64** of the sleeve **20**, as best shown in FIG. **33**. The compression of the suture **30e** between the suture receiving portion **22g** and the tip **32** can fix or lock the suture **30e** to the suture anchor **10g**. Thus, the suture anchor **10g** can fix the second end **304** of the suture **30e** to the suture anchor **10g** without the use of a knot, while providing greater access for the operator to couple the second end **304** of the suture **30e** to the seventh alternative insert **18g** prior to the insertion of the suture anchor **10g** into the pre-drilled hole **14**. The portion of the second end **304** that extends from the pre-drilled hole **14** can then be trimmed if desired (not shown).

(110) With reference to FIG. **34**, an eighth alternative insert **18h** is shown. The eighth alternative insert **18h** can include a suture receiving portion **22h**, the breakaway section **26**, and the end section **24** for use with a suture anchor **10h** substantially similar to the suture anchor **10e** described with regard to FIGS. **21-27**. As the breakaway section **26** and end section **24** of the eighth alternative insert **18h** are substantially similar to the breakaway section **26** and end section **24** of the insert **18**, the breakaway section **26** and end section **24** will not be discussed in detail with regard to the eighth alternative insert **18h**. In addition, as the suture receiving portion **22h** is substantially similar to the suture receiving portion **22e** of the fifth alternative insert **18e**, as discussed with FIGS. **21-27**, only the modifications to the suture receiving portion **22h** will be discussed herein.

(111) As shown in FIG. **34**, the suture receiving portion **22h** can include a hinged slot **370** which can be for receipt of the suture **30e**. Generally, the hinged slot **370** can be defined in a cylindrical body **34h** of the eighth alternative insert **18h**. As will be appreciated, the remainder of the cylindrical body **34h** can be generally similar to the cylindrical body **34e** that is illustrated in and described in conjunction with FIG. **21**.

(112) The hinged slot **370** can define a throughbore **372** and can include a hinged portion **374**. The hinged portion **374** can optionally include at least one or a plurality of barbs or projections **376**. The throughbore **372** can be configured to retain the first end **300** and second end **304** of the suture **30e**. The hinged portion **374** can be pivotably coupled to the cylindrical body **34h** and can rotate from an opened position to a closed position. In the opened position, as shown in FIG. **34**, the hinged portion **374** can define an aperture **378** for receipt of the first end **300** and second end **304** of the suture **30e** to enable the suture **30e** to enter the throughbore **372** of the hinged slot **370**. In the closed position, the hinged portion **374** can substantially enclose the aperture **378** to retain the suture **30e** within the throughbore **372**, as shown in FIG. **35**. Thus, generally, the hinged portion **374** can be sized such that in the closed position the hinged portion **374** can substantially enclose the aperture **378**. The projections **376** are optional and can be coupled to or formed with the hinged portion **374** if desired. The projections **376** can extend into the throughbore **372** when the hinged portion **374** is in the closed position to further retain the suture **30e** within the throughbore **372**. The projections **376** are illustrated as triangular, however, any appropriate shape, such as annular or hooked, could be employed.

(113) As the assembly and deployment of the suture anchor **10h** in the pre-drilled hole **14** is substantially similar to the assembly and deployment of the suture anchor **10e** discussed with regard to FIGS. **21-27** and the suture anchor **10f** discussed with regard to FIGS. **30** and **31**, the assembly and deployment of the suture anchor **10h** will not be discussed in detail herein. Briefly, however, the first end **300** of the suture **30e** can be tied to the breakaway section **26** and the midsection **302** of the suture **30e** can be coupled to the soft tissue **306** via the mattress stitch **308** or any other technique, such as a suture punch (not shown). The hand **86** of the operator can thread or insert the second end **304** of the suture **30e** into the hinged slot **370** (not shown). Then, the suture anchor **10h** can be inserted into the pre-drilled hole **14** and the hand **86** of the operator can be used to tighten the suture **30e**, and thus the soft tissue **306**, to the pre-drilled hole **14** by pulling the second end **304** of the suture **30e** that extends from the pre-drilled hole **14** (not shown). This further removes any slack from the suture **30e**. After the suture **30e** is tightened, the expanding members

72 of the suture anchor **10h** can be deployed to engage the pre-drilled hole **14**. During the deployment of the suture anchor **10h**, as the eighth alternative insert **18h** moves rearwardly in the sleeve **20**, the hinged portion **374** can be pushed closed by the sleeve **20** to lock the suture **30e** within the throughbore **372** of the suture receiving portion **22h**.

(114) When the suture anchor **10h** is fully engaged to the pre-drilled hole **14**, the tip **32** of the suture receiving portion **22h** can be adjacent to the suture **30e**, while the suture **30e** is adjacent to the first end **64** of the sleeve **20**, as best shown in FIG. **35**. The compression of the suture **30e** between the suture receiving portion **22h** and the tip **32** can fix or lock the suture **30e** to the suture anchor **10h**. Thus, the suture anchor **10h** can also fix the second end **304** of the suture **30e** to the suture anchor **10h** without the use of a knot. The portion of the second end **304** that extends from the pre-drilled hole **14** can then be trimmed if desired (not shown).

(115) With reference to FIG. **36**, a ninth alternative insert **18i** is shown. The ninth alternative insert **18i** can include a suture receiving portion **22i**, the breakaway section **26**, and the end section **24** for use with a suture anchor **10i** substantially similar to the suture anchor **10** described with regard to FIGS. **1-7**. As the breakaway section **26** and end section **24** of the ninth alternative insert **18i** are substantially similar to the breakaway section **26** and end section **24** of the insert **18**, the breakaway section **26** and end section **24** will not be discussed in detail with regard to the ninth alternative insert **18i**. In addition, as the suture receiving portion **22i** is substantially similar to the suture receiving portion **22** of the insert **18**, as discussed with FIGS. **1-6B**, only the modifications to the suture receiving portion **22i** will be discussed herein.

(116) As shown in FIG. **36**, the suture receiving portion **22i** includes the eyelet **36** and a hinged slot **400**. Generally, the eyelet **36** and hinged slot **400** can each be defined in a cylindrical body **34i** of the ninth alternative insert **18i**. As will be appreciated, the remainder of the cylindrical body **34i** can be generally similar to the cylindrical body **34** that is illustrated in and described in conjunction with FIGS. **1-6B**. The eyelet **36** can be configured for receipt of the suture **30** therein, as discussed herein, while the hinged slot **400** can be configured for receipt of a suture **30i** therein.

(117) The hinged slot **400** can include a throughbore **402**, a first hinged portion **404** and a second hinged portion **406**. The throughbore **402** defines a passageway for receipt of the suture **30i** therethrough and can include a first end **408** and a second end **410**. The first hinged portion **404** can be pivotably coupled to the first end **408** of the throughbore **402** and the second hinged portion **406** can be pivotably coupled to the second end **410** of the throughbore **402**. The first hinged portion **404** and second hinged portion **406** can generally be pivotable from an opened position to a closed position. In the opened position, as shown in FIG. **36**, the first hinged portion **404** and second hinged portion **406** can enable the suture **30i** to be received into and through the throughbore **402**. In the closed position, as shown in FIG. **37**, the first hinged portion **404** and second hinged portion **406** can generally pivot to enclose the first end **408** and second end **410** of the throughbore **402**, respectively, to secure or lock the suture **30i** within the throughbore **402**. Thus, generally, the first hinged portion **404** and the second hinged portion **406** can be sized to create an interference fit between the suture **30i**, the first and second hinged portion **404**, **406** and the throughbore **402** to lock the suture **30i** to the ninth alternative insert **18i**.

(118) The suture **30i** can include a first end **412**, a midsection **414** and a second end **416**. The suture **30i** can be used to couple multiple suture anchors **10** together, wherein the reference numeral “**10**” denotes all of the suture anchors **10**. Generally, the first end **412** of the suture **30i** can be coupled to an adjacent suture anchor **10** (not specifically shown) and the midsection **414** of the suture **30i** can pass through the throughbore **402** to secure the suture **30i** to the suture anchor **10i**. The second end **416** of the suture **30i** can then be coupled to another suture anchor **10**, or can be trimmed.

Alternatively, the midsection **414** of the suture **30i** can be coupled to additional soft tissue (not shown). If the midsection **414** is coupled to additional soft tissue, then the first end **412** and second end **416** of the suture **30i** can pass through the throughbore **402** to couple the additional soft tissue to the suture anchor **10**, as shown in FIG. **36A**.

(119) As the assembly and deployment of the suture anchor **10i** in the pre-drilled hole **14** is substantially similar to the deployment of the suture anchor **10** discussed with regard to FIGS. **4A-6B** and the suture anchor **10a** discussed with regard to FIGS. **8-12**, the assembly and deployment of the suture anchor **10i** will not be discussed in detail herein. Briefly, however, in order to employ the suture anchor **10i**, the suture **30** can be threaded through the eyelet **36** of the ninth alternative insert **18i** and then the sleeve **20** can be disposed over the ninth alternative insert **18i** (not specifically shown). The suture **30i** can then be coupled to an adjacent suture anchor **10** or to the additional soft tissue via a mattress stitch or any other technique, such as a suture punch (not shown). If the suture **30i** is coupled to an adjacent suture anchor **10**, then the hand **86** of the operator can thread or insert the second end **416** of the suture **30e** into the throughbore **402** until the midsection **414** is received in the throughbore **402** of hinged slot **400**, and the second end **416** of the suture **30i** extends beyond the second end **410** of the throughbore **402**. If the suture **30i** is coupled to the additional soft tissue, then the hand **86** of the operator can thread or insert the first end **412** and the second end **416** of the suture **30i** into the hinged slot **400** (not shown). Once the suture **30i** is threaded through the hinged slot **400**, the suture anchor **10i** can be inserted into the pre-drilled hole **14**.

(120) The hand **86** of the operator can be used to tighten the suture **30i** to the pre-drilled hole **14** by pulling the respective first end **414** and/or second end **416** of the suture **30i** that extends from the pre-drilled hole **14**. This further removes any slack from the suture **30i** (not shown). After the suture **30i** is tightened, the expanding members **72** of the suture anchor **10i** can be deployed to engage the pre-drilled hole **14**. During the deployment of the suture anchor **10i**, as the ninth alternative insert **18i** moves rearwardly in the sleeve **20**, the first hinged portion **404** and second hinged portion **406** can be pushed closed by the sleeve **20** to lock the suture **30i** within the throughbore **402** of the suture receiving portion **22i** as shown in FIGS. **37** and **37B**. In addition, if necessary, the suture **30** coupled to the eyelet **36** can be pulled by the hand **86** of the operator to fully expand the expanding members **72** in soft boney tissue, as discussed herein.

(121) When the suture anchor **10i** is fully engaged to the pre-drilled hole **14**, the tip **32** of the suture receiving portion **22i** can be adjacent to the suture **30i**, while the suture **30i** is adjacent to the first end **64** of the sleeve **20**, as best shown in FIGS. **37** and **37B**. The compression of the suture **30i** between the suture receiving portion **22i** and the tip **32** can fix or lock the suture **30i** to the suture anchor **10i**. Thus, the suture anchor **10i** can fix the first end **412** (FIG. **37**), and the first end **412** and the second end **416** (FIG. **37B**) of the suture **30i** to the suture anchor **10i** without the use of a knot. The portion of the first end **412** and/or second end **416** that extends from the pre-drilled hole **14** can then be trimmed if desired (not shown).

(122) With reference to FIG. **38**, a tenth alternative insert **18j** is shown. The tenth alternative insert **18j** can include a suture receiving portion **22j**, the breakaway section **26**, and the end section **24** for use with a suture anchor **10j** substantially similar to the suture anchor **10** described with regard to FIGS. **1-6B**. As the breakaway section **26** and end section **24** of the tenth alternative insert **18j** are substantially similar to the breakaway section **26** and end section **24** of the insert **18**, the breakaway section **26** and end section **24** will not be discussed in detail with regard to the tenth alternative insert **18j**. In addition, as the suture receiving portion **22j** is substantially similar to the suture receiving portion **22** of the insert **18**, as discussed with FIGS. **1-6B**, only the modifications to the suture receiving portion **22j** will be discussed herein.

(123) As shown in FIG. **36**, the suture receiving portion **22j** includes the eyelet **36** and a second formed eyelet **450**. Generally, the eyelet **36** and second formed eyelet **450** can each be defined in a cylindrical body **34j** of the tenth alternative insert **18j**. As will be appreciated, the remainder of the cylindrical body **34j** can be generally similar to the cylindrical body **34** that is illustrated in and described in conjunction with FIGS. **1-6B**. The eyelet **36** can be configured for receipt of the suture **30** therein, as discussed previously, while the second formed eyelet **450** can be configured for receipt of the suture **30j** therein.

(124) The second formed eyelet **450** can generally be formed perpendicular to the eyelet **36**;

however, the second formed eyelet **450** could be formed in any appropriate orientation with regard to the eyelet **36**, such as parallel. The second formed eyelet **450** defines a passageway for receipt of a suture **30j** therethrough. The suture **30j** can be used to couple multiple suture anchors **10** together, wherein the reference numeral “**10**” denotes all of the suture anchors **10**, or the suture **30j** can be used to couple additional soft tissue to the suture anchor **10j**. The suture **30j** can include a first end **418**, a midsection **420** and a second end **422**. The midsection **420** can be coupled to either the suture anchor **10** or additional soft tissue, while the first end **418** and second end **422** can pass through the second formed eyelet **450** to couple the suture **30j** to the suture anchor **10j**.

(125) As the assembly and deployment of the suture anchor **10j** in the pre-drilled hole **14** is substantially similar to the assembly and deployment of the suture anchor **10** discussed with regard to FIGS. **4A-6B** and the suture anchor **10a** discussed with regard to FIGS. **8-12**, the assembly and deployment of the suture anchor **10j** will not be discussed in detail herein. Briefly, however, in order to employ the suture anchor **10j**, the suture **30** can be threaded through the eyelet **36** of the tenth alternative insert **18j** and then the sleeve **20** can be disposed over the tenth alternative insert **18j** (not specifically shown). The suture **30j** can then be coupled to an adjacent suture anchor **10** or to additional soft tissue via the mattress stitch or any other technique, such as a suture punch (not shown).

(126) If the suture **30j** is coupled to an adjacent suture anchor **10**, then the hand **86** of the operator can thread or insert the second end **416** of the suture **30j** into the second formed eyelet **450** until the midsection **414** is received in the second formed eyelet **450**, and the second end **416** of the suture **30j** extends beyond the suture anchor **10j**. If the suture **30i** is coupled to the additional soft tissue, then the hand **86** of the operator can thread or insert the first end **412** and the second end **416** of the suture **30j** into the second formed eyelet **450** (not shown). Once the suture **30j** is threaded through the second formed eyelet **450**, the suture anchor **10j** can be inserted into the pre-drilled hole **14**.

(127) As discussed herein, the hand **86** of the operator can be used to tighten the suture **30j** to the pre-drilled hole **14** by pulling the first end **418** and second end **416** of the suture **30j** that extends from the pre-drilled hole **14** (not shown). After the suture **30j** is tightened, the expanding members **72** of the suture anchor **10j** can be deployed to engage the pre-drilled hole **14**. It should be noted, however, that if necessary, the suture **30** coupled to the eyelet **36** can be pulled by a hand **86** of an operator to fully expand the expanding members **72**, as discussed herein.

(128) When the suture anchor **10j** is fully engaged to the pre-drilled hole **14**, the tip **32** of the suture receiving portion **22j** can be adjacent to the suture **30j**, while the suture **30j** is adjacent to the first end **64** of the sleeve **20**, as best shown in FIG. **39**. The compression of the suture **30j** between the suture receiving portion **22j** and the tip **32** can fix or lock the suture **30j** to the suture anchor **10j**. Thus, the suture anchor **10j** can fix the first end **412** and second end **416** of the suture **30j** to the suture anchor **10j** without the use of a knot. The portion of the first end **412** and second end **416** of the suture **30j** that extends from the pre-drilled hole **14** can then be trimmed if desired (not shown).

(129) With reference to FIG. **40**, an eleventh alternative insert **18k** is shown. The eleventh alternative insert **18k** can include a suture receiving portion **22k**, the breakaway section **26**, and the end section **24** for use with a suture anchor **10k** substantially similar to the suture anchor **10e** described with regard to FIGS. **21-27**. As the breakaway section **26** and end section **24** of the eleventh alternative insert **18k** are substantially similar to the breakaway section **26** and end section **24** of the insert **18**, the breakaway section **26** and end section **24** will not be discussed in detail with regard to the eleventh alternative insert **18k**. In addition, as the suture receiving portion **22k** is substantially similar to the suture receiving portion **22e** of the fifth alternative insert **18e**, as discussed with FIGS. **21-27**, only the modifications to the suture receiving portion **22k** will be discussed herein.

(130) As shown in FIG. **40**, the suture receiving portion **22k** can include the eyelet **36** and a second formed eyelet **460**. Generally, the eyelet **36** and second formed eyelet **460** can each be defined in a cylindrical body **34k** of the eleventh alternative insert **18k**. As will be appreciated, the remainder of

the cylindrical body **34k** can be generally similar to the cylindrical body **34** that is illustrated in and described in conjunction with FIGS. **1-6B**. The eyelet **36** can be configured for receipt of a suture **30'** therein, while the second formed eyelet **460** can be configured for receipt of the suture **30k** therein. The suture **30'** can include a first end **454**, a midsection **456** and second end **458**. The first end **454** of the suture **30'** can be coupled to a suture anchor **10** or additional soft tissue **462** and can pass through the second formed eyelet **460** to couple the suture **30'** to the suture anchor **10k**. The midsection **456** can be received through the eyelet **36** to further retain the suture **30'** within the suture anchor **10k**. The second end **458** can extend from the second end **66** of the sleeve **20**.

(131) The second formed eyelet **460** can generally be formed perpendicular to the eyelet **36**; however, the second formed eyelet **460** could be formed in any appropriate orientation with regard to the eyelet **36**, such as parallel. The second formed eyelet **460** defines a passageway for receipt of the suture **30k** therethrough. The suture **30k** can be used to couple multiple suture anchors **10** together, wherein the reference numeral "**10**" denotes all of the suture anchors **10**, or the suture **30k** can be used to couple additional soft tissue **462** to the suture anchor **10k**. The suture **30k** can include a first end **464**, a midsection **466**, and a second end **468**. The first end **464** can be coupled to the breakaway section **26** of the eleventh alternative insert **18k**. The midsection **466** can be coupled to either the suture anchor **10** or to the additional soft tissue **462**. The second end **468** of the suture **30k** can pass through the second formed eyelet **460** to couple the suture **30k** to the suture anchor **10k**.

(132) Typically, in order to employ the suture anchor **10k**, with additional reference to FIG. **41**, the suture **30'** can be threaded through the eyelet **36** of the eleventh alternative insert **18k** and then the first end **412** of the suture **30k** can be tied to the breakaway section **26**. The sleeve **20** can be disposed over the eleventh alternative insert **18k** such that the suture **30'** and the suture **30k** extend from the second end **66** of the sleeve **20**. Then, the midsection **466** of the suture **30k** and a portion **470** of the first end **454** of the suture **30'** can then be coupled to an adjacent suture anchor **10** (not shown) or to the additional soft tissue **462** via the mattress stitch **472** or any other technique, such as a suture punch. If the suture **30'** and the suture **30k** are coupled to an adjacent suture anchor **10**, then the hand **86** of the operator can thread or insert the first end **454** of the suture **30'** and the second end **468** of the suture **30k** into the second formed eyelet **460** until the first end **454** of the suture **30'** and the second end **468** of the suture **30k** extend beyond the suture anchor **10k** (not shown).

(133) If the midsection **466** of the suture **30k** and the portion **470** of the first end **454** of the suture **30'** are coupled to the additional soft tissue **462**, then the hand **86** of the operator can thread or insert the first end **454** of the suture **30'** and the second end **468** of the suture **30k** into the second formed eyelet **460**, as shown in FIG. **41**. Once the first end **454** of the suture **30'** and the suture **30k** are threaded through the second formed eyelet **460**, the suture anchor **10k** can be inserted into the pre-drilled hole **14**.

(134) The hand **86** of the operator can be used to tighten the additional soft tissue **462** to the pre-drilled hole **14** by pulling the suture **30k** and the first end **454** of the suture **30'** that extends from the pre-drilled hole **14**. After the suture **30'** and suture **30k** are tightened, the expanding members **72** of the suture anchor **10k** can be deployed to engage the pre-drilled hole **14**. As the deployment of the suture anchor **10k** in the pre-drilled hole **14** is substantially similar to the deployment of the suture anchor **10** discussed with regard to FIGS. **4A-6B** and the suture anchor **10e** discussed with regard to FIGS. **21-27**, the deployment of the suture anchor **10k** will not be discussed in detail herein. It should be noted, however, that if necessary, the first end **454** and second end **458** of the suture **30'** coupled to the eyelet **36** can be pulled by the hand **86** of the operator to fully expand the expanding members **72** in the case of soft boney tissue, as discussed herein.

(135) When the suture anchor **10k** is fully engaged to the pre-drilled hole **14**, the tip **32** of the suture receiving portion **22k** can be adjacent to the suture **30'** and the suture **30k**, while the suture **30'** and suture **30k** are adjacent to the first end **64** of the sleeve **20**, as best shown in FIG. **42**. The

compression of the suture **30'** and the suture **30k** between the suture receiving portion **22k** and the tip **32** can fix or lock the suture **30'** and the suture **30k** to the suture anchor **10k**. Thus, the suture anchor **10k** can fix the suture **30'** and the suture **30k** to the suture anchor **10k** without the use of a knot. The portion of the second end **458** of the suture **30'** and the suture **30k** that extend from the pre-drilled hole **14** can then be trimmed if desired (not shown).

(136) The description of these teachings is merely exemplary in nature and, thus, variations that do not depart from the gist of the teachings are intended to be within the scope of the teachings. Such variations are not to be regarded as a departure from the spirit and scope of the teachings.

Claims

1. A suture anchor system for anchoring soft tissue to bone, the suture anchor system comprising: a suture anchor including a sleeve anchor and a suture-receiving insert, the sleeve anchor including an interior side wall and exterior threads, the suture-receiving insert including a distal portion and a proximal body portion that extends proximally of the distal portion, the distal portion including a distal tip and a proximal end with a proximal facing surface; and a suture passable through an eyelet in the suture-receiving insert and couplable to soft tissue, wherein the suture-receiving insert is partially receivable in the sleeve anchor to lock the suture to the suture anchor, said partially receivable including the proximal body portion extending into the sleeve anchor through an opening in a distal end of the sleeve anchor with the distal portion remaining outside the sleeve anchor distal of the opening in the distal end of the sleeve anchor, wherein the distal end of the sleeve anchor includes a distal facing surface around the opening, and wherein the suture being so locked includes the suture extending from a first free end of the suture distally along the proximal body portion between an exterior side wall of the proximal body portion and the interior side wall of the sleeve anchor, followed by the suture exiting and subsequently re-entering the suture anchor by passing a first time and a second time, respectively, between the proximal facing surface of the proximal end of the distal portion and the distal facing surface of the distal end of the sleeve anchor without passing through the sleeve anchor, followed by the suture passing through the eyelet in the suture-receiving insert, followed by the suture re-exiting the suture anchor by passing a third time between the proximal facing surface of the proximal end of the distal portion and the distal facing surface of the distal end of the sleeve anchor, wherein the first free end of the suture is wholly contained within the sleeve anchor when the suture is locked to the suture anchor.
2. The suture anchor system of claim 1, wherein the first free end of the suture is fixedly coupled to the proximal body portion of the suture-receiving insert.
3. The suture anchor system of claim 1, wherein a diameter of the proximal body portion is such that an interference fit occurs between the proximal body portion, the suture, and the sleeve anchor.
4. The suture anchor system of claim 1, wherein the suture is lockable to the suture anchor without the use of a knot.
5. The suture anchor system of claim 1, wherein the distal facing surface of the sleeve anchor occurs at the distal end of the sleeve anchor.
6. The suture anchor system of claim 2, wherein the first free end of the suture is knotted to the proximal body portion of the suture-receiving insert.
7. A suture anchor system for anchoring soft tissue to bone, the suture anchor system comprising: a suture anchor including a sleeve anchor and a suture-receiving insert, the sleeve anchor including a distal end opening and a proximal end opening, the sleeve anchor further including an interior side wall and exterior threads, the suture-receiving insert including a distal portion and a proximal body portion that extends proximally of the distal portion, the distal portion including a distal tip and a proximal end with a proximal facing surface; and a suture passable through an eyelet in the suture-receiving insert and couplable to soft tissue, wherein the suture-receiving insert is partially receivable in the sleeve anchor to lock the suture to the suture anchor, said partially receivable

including the proximal body portion extending into the sleeve anchor through the distal end opening in the sleeve anchor with the distal portion remaining outside the sleeve anchor distal of the distal end opening, wherein the sleeve anchor includes a distal facing surface around the distal end opening, and wherein the suture being so locked includes the suture extending from a first free end of the suture that is positioned inside the sleeve anchor distally along the proximal body portion between an exterior side wall of the proximal body portion and the interior side wall of the sleeve anchor to exit the sleeve anchor through the distal end opening without ever passing through the proximal end opening in the sleeve anchor, followed by the suture exiting and subsequently re-entering the suture anchor by passing a first time and a second time, respectively, between the proximal facing surface of the proximal end of the distal portion and the distal facing surface of the sleeve anchor without passing through the sleeve anchor, followed by the suture passing through the eyelet in the suture-receiving insert, followed by the suture re-exiting the suture anchor by passing a third time between the proximal facing surface of the proximal end of the distal portion and the distal facing surface of the distal end of the sleeve anchor, wherein the first free end of the suture is fixedly coupled to the suture-receiving insert independent of the sleeve anchor.

8. The suture anchor system of claim 7, wherein the distal tip is conical.

9. The suture anchor system of claim 7, wherein the proximal body portion is cylindrical.

10. The suture anchor system of claim 7, wherein the proximal body portion is slidably receivable in the sleeve anchor.

11. The suture anchor system of claim 7, wherein the first free end of the suture is wholly contained within the sleeve anchor when the suture is so locked to the suture anchor.

12. The suture anchor system of claim 7, wherein a diameter of the proximal body portion is such that an interference fit occurs between the proximal body portion, the suture, and the sleeve anchor.

13. The suture anchor system of claim 7, wherein the first free end of the suture is knotted to the proximal body portion of the suture-receiving insert.

14. The suture anchor system of claim 7, wherein the distal facing surface of the sleeve anchor occurs at the distal end of the sleeve anchor.

15. The suture anchor system of claim 7 further comprising a suture anchor implanting tool removably engaging the suture-receiving insert and the sleeve anchor.

16. The suture anchor system of claim 15, wherein the suture anchor implanting tool is configured to deliver the suture-receiving insert into the bone ahead of the sleeve anchor.

17. A suture anchor system for anchoring soft tissue to bone, the suture anchor system comprising: a suture anchor including a sleeve anchor and a suture-receiving insert, the sleeve anchor including a distal end opening and a proximal end opening, the sleeve anchor further including an interior side wall and exterior threads, the suture-receiving insert including a distal portion and a proximal body portion that extends proximally of the distal portion, the distal portion including a distal tip and a proximal end with a proximal facing surface; and a suture passable through an eyelet in the suture-receiving insert and couplable to soft tissue, wherein the suture-receiving insert is partially receivable in the sleeve anchor to lock the suture to the suture anchor, said partially receivable including the proximal body portion extending into the sleeve anchor through the distal end opening in the sleeve anchor with the distal portion remaining outside the sleeve anchor distal of the distal end opening, wherein the distal end of the sleeve anchor includes a distal facing surface around the distal end opening, and wherein the suture being so locked includes the suture extending from a first free end of the suture that is positioned inside the sleeve anchor distally along the proximal body portion between an exterior side wall of the proximal body portion and the interior side wall of the sleeve anchor to exit the sleeve anchor through the distal end opening without ever passing through the proximal end opening in the sleeve anchor, followed by the suture exiting and subsequently re-entering the suture anchor by passing a first time and a second time, respectively, between the proximal facing surface of the proximal end of the distal portion and the distal facing surface of the distal end of the sleeve anchor without passing through the sleeve anchor, followed

by the suture passing through the eyelet in the suture-receiving insert, followed by the suture re-exiting the suture anchor by passing a third time between the proximal facing surface of the proximal end of the distal portion and the distal facing surface of the distal end of the sleeve anchor, followed by the suture extending to a second free end of the suture positioned proximally away from a proximal end of the sleeve anchor by extending outside the sleeve anchor proximally along the exterior threads of the sleeve anchor.

18. The suture anchor system of claim 17, wherein the first free end of the suture is fixedly coupled to the suture-receiving insert independent of the sleeve anchor.

19. The suture anchor system of claim 17, wherein the first free end of the suture is wholly contained within the sleeve anchor when the suture is locked to the suture anchor.

20. The suture anchor system of claim 17, wherein the proximal body portion is slidably receivable in the sleeve anchor.

21. The suture anchor system of claim 17, wherein the distal facing surface of the sleeve anchor occurs at the distal end of the sleeve anchor.
